Beyond Factual QA: Extending the Low-Resource Hallucination-Detector Robustness Phenomenon to Dialogue Generation and a Broader Set of Languages

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Model selection

All experiments use one generator, meta-llama/Llama-3.1-8B-Instruct, loaded in 4-bit NF4 with bitsandbytes and bfloat16 compute. The proposal fixed this choice because the model fits a 16 GB GPU after quantization, and one model everywhere keeps our mTREx run comparable with X-FACTR and PsiloQA. Access to the gated weights was granted, so the Mistral fallback was not needed. The run used one NVIDIA RTX A4000 (16 GB) and peaked at 5.73 GiB of GPU memory.

The three detectors come from the authors' released code (Datta et al., EACL 2026), pinned at commit 644d655:

- **MAM**: a small classifier (one hidden layer of 256 units) trained on the model's internal activations at the first generated token, from the final layer's fully connected block (FC) and its self-attention output (Attn).
- **SelfCheckGPT**: draws 20 extra answers and scores how poorly they support the main answer (unigram score).
- **Semantic Entropy**: groups the 20 answers by meaning with LaBSE embeddings (cosine threshold 0.75) and measures how spread out the groups are.

Implementation

We copied the authors' scripts into src/detectors and changed only what our setting needs: 4-bit loading on one GPU, loaders for our record format, labels from our matches() function, prompts for the new languages, resume after a crash and a log of failed items. Each change carries a G6 marker, a saved diff and an entry in src/detectors/AUTHORS_CODE.md, and a test fails if any unmarked function differs from the original. Two changes affect what is stored:

- **One shared answer**: each item gets one greedy answer (5 to 50 new tokens, as in the released MAM script), and all three detectors score that text. The released sampling scripts

each make their own answer, which would give one item two labels.

- **Entropy instead of logits**: we store the entropy of the first token's probability distribution, not all 128,256 values, which is what the Phase 5 entropy check needs.

The 20 samples per item are drawn once at temperature 1 (top_p 0.9 from the model's generation settings) and shared by both sampling detectors. A final check confirms that every output file matches the processed data, has no duplicates, and has both classes in each test split.

Pipeline

One script, scripts/run_phase3.py, runs a smoke test on 20 items per dataset, a timing test that set the run sizes in configs/run.yaml, the full run and the final check. For every dataset and language it generates the greedy answer and saves the activations MAM needs, labels the answer, trains the MAM classifiers, and scores the first 91 items in hash order with the two sampling detectors. On mTREx and X-FACTR an answer is hallucinated when it contains none of the accepted answers after normalisation; on PsiloQA the label is the human span annotation. PsiloQA answers were written by other models, so MAM reads our model's activations over the given answer's tokens (the authors' multi-token variant).

The run covered 47,077 items: mTREx (5 languages, 32,100 items), X-FACTR (8 languages including English, 9,977 items) and PsiloQA (5 languages, 5,000 items). It took about 16 hours from the start of the full run to the final check, and no item failed.

Training process

Only MAM is trained. Within each language we split the items 80/20 at random and train on the 80% with the authors' settings: AdamW, learning rate 1e-4, weight decay 1e-2, batch size 128, about 1,000 steps, and one classifier per question category on mTREx. We repeat the split with seeds 42, 43 and 44, so each MAM AUROC below is a mean over three held-out sets. The training seed is fixed too, so a rerun gives the same numbers.

Initial results

Table 1 gives accuracy (the share of correct answers) and AUROC per detector, where 0.5 is chance and 1 separates correct and hallucinated answers perfectly. Confidence intervals, drops relative to English and TPHR come in Phase 4.

Table 1. Accuracy and detector AUROC per language. MAM is the mean over three splits. SelfCheckGPT and Semantic Entropy use 91 items per language.

Dataset	Language	Items	Accuracy	MAM (FC)	MAM (Attn)	SelfCheckGPT	Sem. Entropy
mTREx	English	6,420	69.5%	0.83	0.85	0.53	0.79
mTREx	German	6,420	61.6%	0.84	0.84	0.68	0.59
mTREx	Hindi	6,420	35.4%	0.78	0.78	0.58	0.68
mTREx	Bengali	6,420	29.1%	0.81	0.81	0.61	0.65
mTREx	Urdu	6,420	30.2%	0.74	0.76	0.74	0.61
X-FACTR	English	1,796	51.3%	0.81	0.80	0.52	0.77
X-FACTR	Chinese	1,380	38.6%	0.80	0.80	0.54	0.69
X-FACTR	Japanese	1,380	29.8%	0.84	0.83	0.73	0.75
X-FACTR	Vietnamese	1,380	41.3%	0.79	0.81	0.62	0.80
X-FACTR	Tagalog	1,215	30.0%	0.82	0.82	0.48	0.70
X-FACTR	Swahili	1,136	36.2%	0.73	0.78	0.73	0.73
X-FACTR	Yoruba	856	7.4%	0.79	0.78	0.53	0.73
X-FACTR	Ilocano	834	27.5%	0.86	0.88	0.66	0.69
PsiloQA	English	1,000	4.3%	0.55	0.56	0.51	0.34
PsiloQA	Hindi	1,000	13.2%	0.59	0.60	0.39	0.29
PsiloQA	Chinese	1,000	13.0%	0.59	0.60	0.56	0.53
PsiloQA	Finnish	1,000	16.0%	0.56	0.64	0.43	0.44
PsiloQA	Arabic	1,000	11.8%	0.71	0.72	0.37	0.44

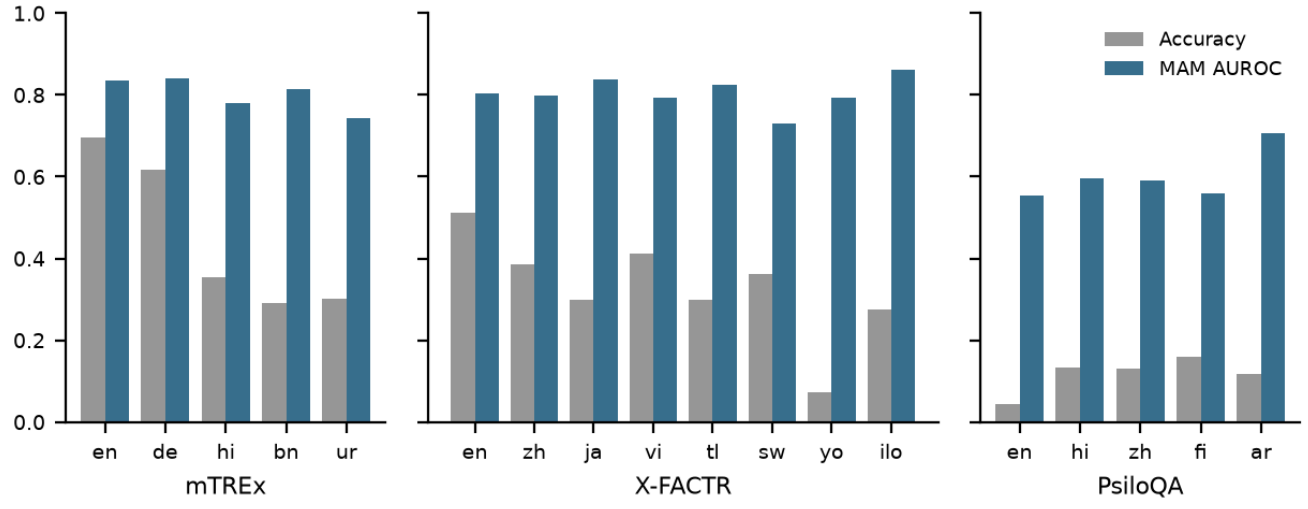


Figure 1. Accuracy and MAM (FC) AUROC per language for each dataset.

- On mTREx, accuracy falls from 70% in English to as low as 29%, while MAM AUROC stays between 0.74 and 0.84, which matches the base paper's finding with our quantized model.
- X-FACTR shows the same pattern over a wider language set. Yoruba answers only 7% correctly, yet its MAM AUROC is 0.79 (English 0.81), and Ilocano reaches 0.86.
- On PsiloQA all detectors are weaker. MAM AUROC ranges from 0.55 to 0.71, and 7 of the 10 sampling scores fall below 0.5. Accuracy there is only 4% to 16%, because most PsiloQA answers contain a hallucinated span.

We checked 400 X-FACTR answers by hand, 50 per language in hash order, and compared each decision with the matcher. Agreement: Japanese 90%, English 88%, Swahili 84%, Yoruba 84%, Ilocano 82%, Chinese 80%, Tagalog 74%, Vietnamese 70%. The matcher rejected 57 correct answers, mostly true answers missing from the gold list for relations with several answers (diplomatic relations, borders, World Cup teams) or translated names, and accepted 17 wrong ones. It counts 36% of the sample as correct against 46% by hand, so X-FACTR accuracy is somewhat understated. Agreement is under 90% in 7 languages, so, as the proposal requires, we add a second label that also accepts an answer equal to a gold alias after stemming. It raises accuracy by at most 0.1 points, since most misses come from incomplete gold lists, which stemming cannot fix. Phase 4 reports both rules.

Challenges

- **Machine shutdown**: the GPU machine switched off during the first long run. We made every output safe against a sudden stop by flushing each line to disk and replacing configuration files in one step. After that the run could restart by itself at login, and it pushed its results to GitHub every hour.
- **Failed pushes**: for several hours the hourly pushes were rejected because the GPU machine was signed in to a GitHub account without access to the repository. The results stayed safe locally until the account was fixed.
- **Class imbalance**: PsiloQA English has only 4% correct answers, so its test splits hold very few correct items (8 and 9 in the seed 43 and 44 test splits) and its AUROC is unstable.
- **Sampling cost**: 20 samples per item are slow, so the sampling detectors ran on 91 items per language, as the proposal's cut order allows. MAM ran on every item.

Code, results and notebook (tag phase3): https://github.com/ahmad1khan2/f26-g6-hallucination-detectors